HANDHELD CRANK PHONE CHARGER

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Texas Tech University

ENGR-1340-012

Introduction to Engineering Design

Ebubechukwu Iwunnah

Quoc Thinh Tran

Steven Finley

Joshua Garcia

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1. INTRODUCTION

This document is the report for a handheld phone charger, a.k.a. HPC. This project proposal will outline our objective, description, environmental and operational scenarios, and project organisation for the device.

1.0 HPC System Concept Description

Mobile phones are one of the most used devices in the world. Being able to communicate with others, enjoy entertainment, learn new skills and knowledge, and find news on current events all in a single device has been revolutionary. Additionally, as opposed to other electronics such as computers, the portability of phones is unmatched.

However, there is one glaring issue with phones: their battery life. A phone must be fully charged every 12 to 24 hours, and this time reduces with prolonged active use. Currently, users use a charger and outlet or a portable charger to keep their device alive. However, these options are not always available in remote areas, during travel, or in emergencies. This then renders the phone useless, with no form of functionality.

To address this, our group plans to find a solution using a charger that can charge a phone wherever you are. More specifically, our group plans to build a portable phone charger that you yourself put energy into to charge, rather than a battery. This will provide a reliable source of power anytime and anywhere.

1.1 HPC Objective

Our group submits this proposal for the concept of a system for a “Handheld Phone Charger.” In this day and age, where more people use their phones, people look for ways to keep their devices at a higher percentage for a longer time. In such cases, a person might bring a phone charger with them to plug into a wall socket. Alternatively, a person might bring along a portable battery bank to charge their phone with. However, if they are in an environment without easily accessible outlets or if their battery bank is dead, they are out of luck. Additionally, many people live in urban environments that push walking or other person-based transportation, which eliminates the need for cars. Thus, people look towards solutions that will not only be consistent but also minimally invasive. If their charger takes up all the space in their pocket or if it weighs too much, it would not be worth the hassle to bring around. Focusing on these, the project report will address the needs of cellphone users in environments where charging their phones is difficult. Additionally, we plan to have the charger track the outputted voltage and the user’s rpm while in use.

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1.2 HPC Needs

To ensure that a product that consumers will enjoy is created, we decided on requirements for the charger that were needed to ensure satisfaction. Below is a summary of the requirements.

- The HPC shall fit dimensions of 6 in x 3 in x 1.5 in
- The HPC shall emit a consistent 5V output
- The HPC shall be comfortable for the user
- Resistance to external trauma
- Accurately displays the output voltage to the user
- Accurately displays the input voltage by the user.

During its development, these needs will be addressed. The charger needs to be able to fit in the user's pockets; otherwise, its portability wouldn't be satisfactory. A user should feel that their effort is worthwhile, not that they're spending excess energy for only a tiny bit of charge. Not only that, but it should also be easy to handle to make the actual charging process easy. Additionally, it should have a Screen able to display the charger's state and values. Finally, it should be able to resist realistic external trauma.

1.3 Environment

While not in use, the HPC will be in two places: the user's pocket or the user's backpack. It is reasonable to assume that the HPC will share space with other objects such as keys, bottles, or school supplies. Although it would be smart to keep it in a separate compartment, the user throwing the HPC into the bag randomly into any of these is the most likely.

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2.0 OPERATIONAL SCENARIOS

2.1 Three Scenarios

We determined three scenarios for which the charger will be used

1) A student is taking it with them to class
2) A tourist taking it with them around the city
3) A person in an unpopulated area (forest or another stressful environment)

The HPC is best utilised under the situation of having no easy access to charging; thus, it will be placed under different scenarios to understand the challenges of each scenario and to understand the customer base. First, a scenario where the HPC is placed into a backpack, then pulled out to be used on a surface. The next scenario is pulling the charger out of a pocket and using it with no surface support. Third, the HPC is dropped, and the user has slippery hands and needs to use the charger.

2.2.1 Scenario 1: Sit Down at a College Campus

2.2.1.1 Overall Description

A new student is walking around campus. They can find any chargers nearby, and their phone is about to die. They then remember that they had their HPC in their backpack. They sit down at a nearby bench and place the HPC on the table, and prepare to charge.

2.2.1.2 Mission Description

The system's objective in this is to be able to charge the user's device in a comfortable environment. The user will place the HPC on the table, connect their charging cord, then start turning the HPC's crank and charging their device. The system is successful if the phone is charging quickly after cranking has started.

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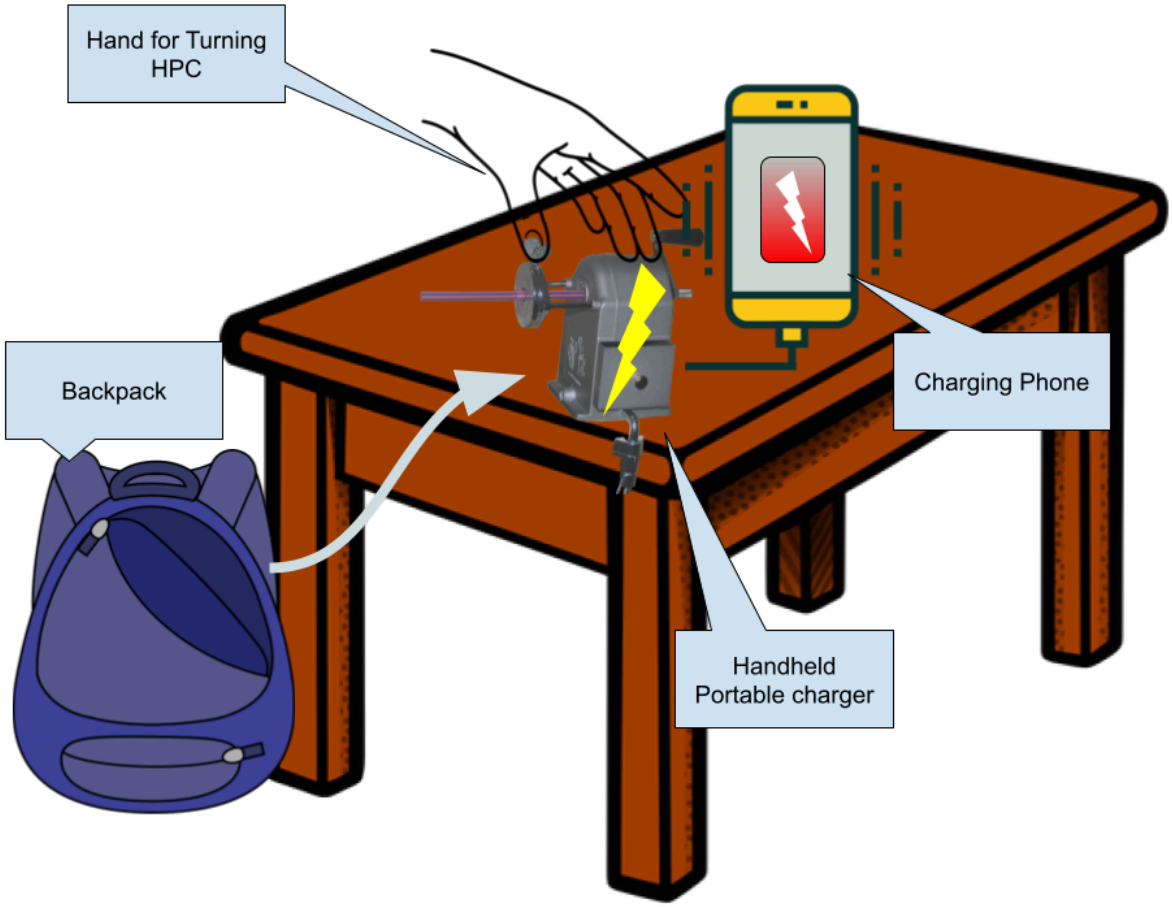


Figure 1: Scenario 1 of HPC, charging on a table.

2.2.1.3 Operational Environment

The system provides a consistent charging experience while being on a stable surface.

2.2.1.4 Physical Environment

The system must be sturdy enough to prevent snapping, breaking, and/or any mishaps while inside the backpack. Additionally, the System should operate within a moderate climate and weather (We will assume 65 – 75 degrees Fahrenheit on a partly cloudy day).

2.2.1.5 Friendly Systems

Different charger coords and surfaces are the Friendly Systems that the HPC will interact with. Phone charger cords are delicate and tend to break easily; thus, the HPC must be compatible with any USB-A male plugs. This ensures that any standard cord of any length and head will be compatible. Additionally, when using a table as a support, the HPC slipping will hinder the user's ability to turn it; thus, some sort of rubber feet or pad should be on the bottom to ensure sturdiness.

2.2.1.6 Threat System

The threats are the various objects in the user's bag. If the HPC can't withstand general shuffling and shaking inside a backpack, it will be fragile for its portability services.

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2.2.2 Standing Up in a Busy City

2.2.2.1 Overall Description

A tourist is visiting {insert generic city}. They're going around enjoying the sights and taking photos and videos of everything they see. At one point, their phone had gotten low on charge and grabbed their HPC out of their pocket, and looked for a table to charge it. However, they couldn't find a table anywhere; thus, they decided to try to use the HPC while standing up.

2.2.2.2 Mission Description

The system's objective in this scenario is to provide a comfortable user experience while in their pocket, doing semi-active activities, proving its tensile strength by resisting bending while in the pocket, and being ergonomic enough to provide a satisfactory charging experience, even when only support is the user's hand/arm. The HPC will be successful if it can reach 5V of charge while being held in the hand.

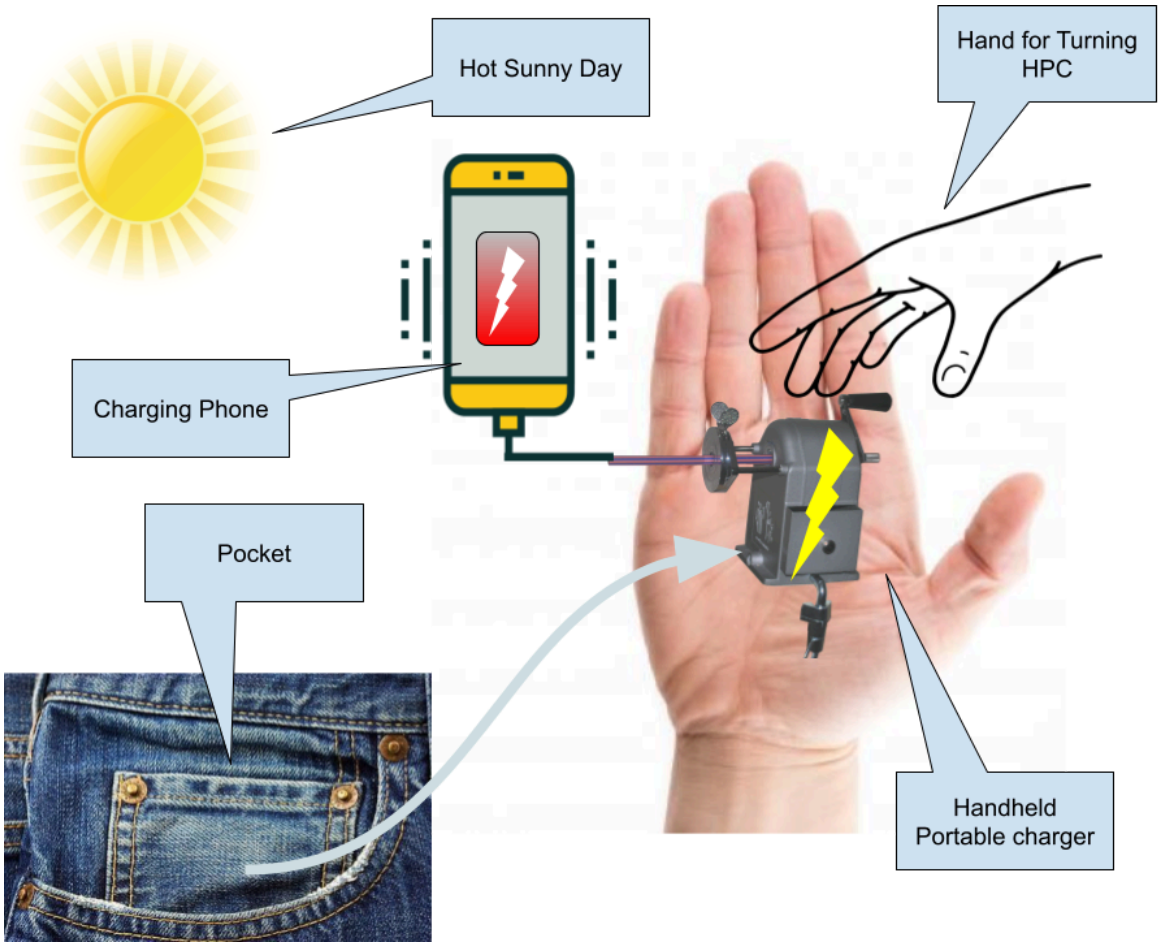


Figure 2: Pulling the Charger out of the pocket to use only supported by the hand

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2.2.2.3 Operational Environment

The HPC needs to be comfortable enough for someone to walk around with. The person should be able to walk normally and feel little to no discomfort. Additionally, the HPC should be durable enough to withstand bending or stretching from being in the user's pocket. There should be no worry of components getting unaligned or snapping occurring. Finally, the HPC should provide an easy-to-use system that, despite not being on a level and solid surface, should still be easy to handle and use.

2.2.2.4 Physical Environment

Not only should the system be operational in safe environments, but the system should also be resistant in tight/ compressive environments, such as a person's pocket. Additionally, the HPC should be able to handle hotter temperatures (85-100 degrees Fahrenheit). These will mimic a typical day of touring during the summer.

2.2.2.5 Friendly Systems

Different charger coords and surfaces are the Friendly Systems that the HPC will interact with. Phone charger cords are delicate and tend to break easily; thus, the HPC must be compatible with any USB-A male plugs. This ensures that any standard cord of any length and head will be compatible. Additionally, the HPC will need to be stable when held in the hands. This will be done using an easy-to-grip design.

2.2.2.6 Threat Systems

Stated numerous times, the biggest threat to the system is not cracking, snapping, or splitting while under tensile stress. Not only will holding the HPC in your hand and turning it cause stress, but it being also in your pocket will induce stress. The HPC will need to be able to resist this stress to be satisfied.

2.2.3 Scenario 3: Emergency

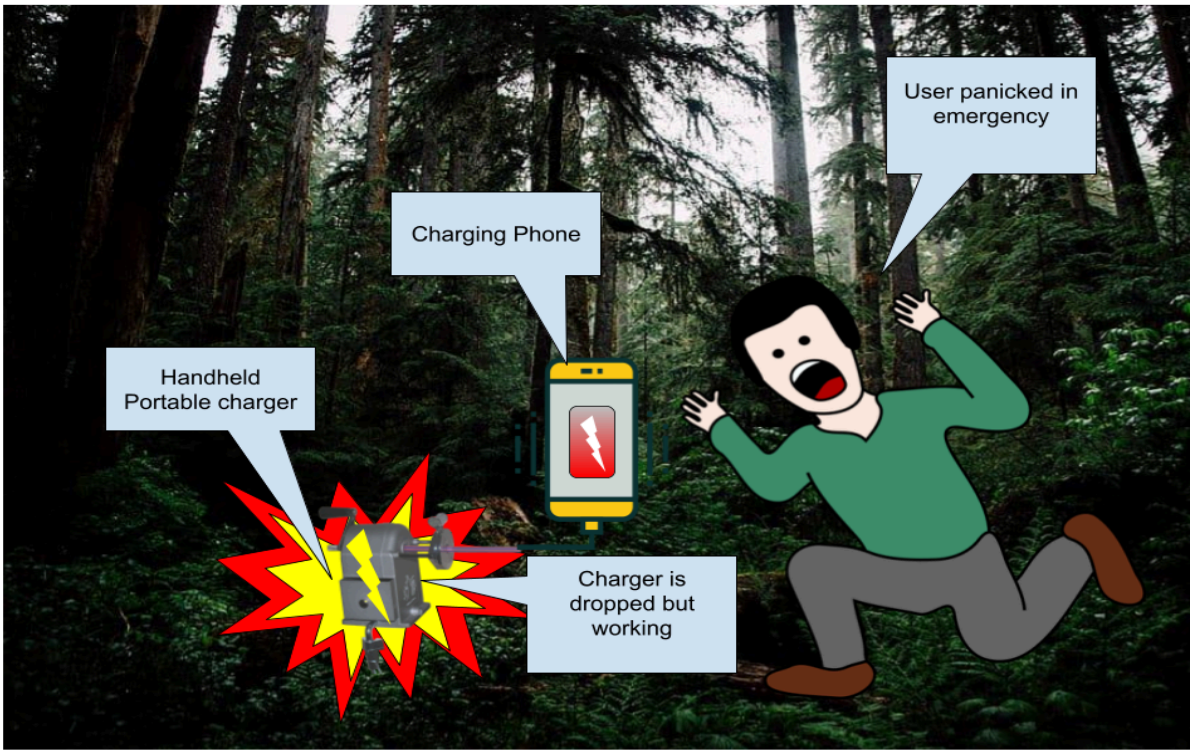
2.2.3.1 Operational Scenario

While in the wilderness, an accident occurred that required Emergency Services. The user's phone is dead, and all they have is HPC. It's dark, and it's hard for them to grab the HPC, so it is dropped repeatedly before use. Then they need their phone charged enough to contact emergency services as fast as possible, and to stay on call for around 5 minutes to have their location and identity taken. The users will also want to know their voltage output to know if they need to spin faster or if their charger is no longer operational.

2.2.3.2 Mission Description

HPC's objective in this scenario is to provide the same charging experience despite being dropped a multitude of times. To survive the falls, a durable design must be made; however, it should still comply with the parameters explored before. Both the externals and internals must be secure to prevent breakage and components from getting loose after the falls. The HPC will be successful if it can still charge, reach, and display 5V of charge after being dropped numerous times.

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2.2.3.3 Operational Environment

The HPC will be placed on the ground and used to charge a phone. The crank will be spun as fast as humanly possible charge will be as fast as it can. The charging will commence after it has been dropped numerous times.

2.2.3.4 Physical Environment

The HPC must provide a consistent charging experience despite being in cold conditions. (40 – 60 degrees Fahrenheit) It must be resistant to having dirt and debris enter the system, as it is assumed to be in some sort of forest or camping trail.

2.2.3.5 Friendly Systems

The HPC will need to work in tandem with the user and the ground. The user will want a consistent charging experience, and the ground will provide an uneven yet stable working surface.

2.2.3.6 Threat System

The HPC will need to be resistant to high-impact trauma and debris. After being dropped numerous times, the biggest issue will arise when components become loose or break. If the internals become compromised or don't work, the user won't be able to charge their device. Additionally, after being dropped, it wouldn't be out of the possibilities for dirt or other debris to generate on the HPC; thus, it has to be able to work even if there is a little bit of mess on it, or be easy to clean so the user can remove the dirt.

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3. System Concept Description

This report describes the system concept for HPC using a context diagram, scenarios, and concept of operations illustrations, consistent with the Kossiakoff systems engineering method.

3.1 HPC System Context Diagram

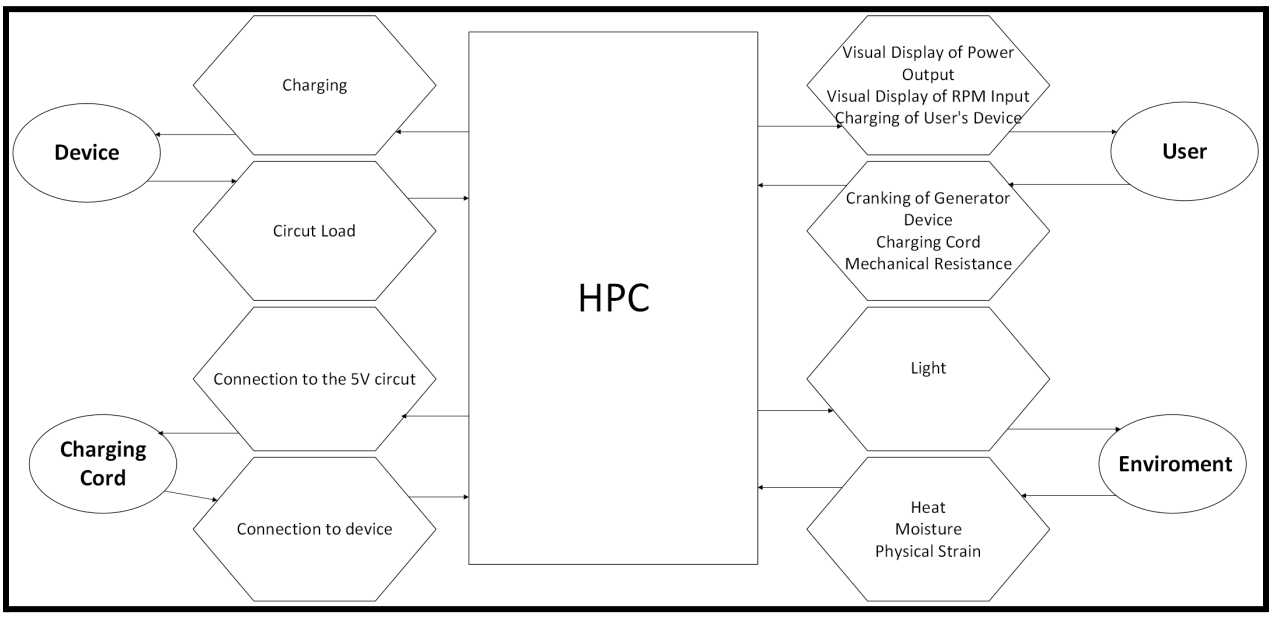
The context diagram for the HPC simply shows how the charger fits into the real world and who or what it interacts with. Each external actor gets its own block: Device, Charging Cable, User, and Environment.

We kept the diagram deliberately implementation-neutral. In practice, that means we treated the charging cable separately from the electronic load (the device being charged), so the diagram doesn't imply a single connector type or one specific way to hook things up.

A few notes on what each block means and why it's there:

- Device — the thing that receives energy. The charger must handle charging, carry the circuit load, and connect to the device's 5 V circuit.
- Charging Cable — modelled on its own because it's a physical link that can wear out or add contact resistance. The diagram shows how it connects both to the device and to the 5 V circuit.
- User — the human who provides mechanical input (cranking the generator), watches the output (power/RPM displays), and ultimately benefits from the charge.
- Environment — all non-human influences like heat, moisture, light, or physical shocks. These factors affect how well the system performs and how reliable it is.

We grouped similar hazards under "Environment" on purpose. That keeps the diagram high-level rather than turning it into a long checklist of every possible failure mode. The interactions drawn from the system description emphasise the interfaces the HPC must satisfy — electrical, mechanical, human-machine, and environmental — so the design team knows what to focus on.



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4. Operational Requirements

4.1 Purpose

This section guides the design so the HPC can reliably charge common mobile devices without an ideal source of power. The ideal purpose would be camping, being lost in the jungle, or any unfortunate situation that may occur.

1. Provide phone charging without access to wall power or batteries.
2. Deliver a stable, USB-compatible 5 V output while cranking.
3. Be compact, lightweight, and easy to carry and use
4. Offer clear feedback (voltage and crank RPM) so users can charge effectively.
5. Operate safely and withstand everyday impacts, drops, and environmental conditions.
6. Work with standard phone cables and devices (broad compatibility)
7. Keep costs reasonable for students and outdoor users.
8. Support sustainable, human-powered operation with minimal maintenance.
9. Comply with applicable electrical, safety, and materials guidelines.

4.2 HPC Operational Requirements

General

1. The charger shall work using only hand cranking (no wall power required)
2. The charger shall start delivering power within 5 seconds of cranking.

Size & Weight

3. The device shall fit within 6 × 3 × 1.5 in when stored.
4. The device shall weigh ≤ 350 g (0.77 lb).

Power & Electrical

5. Output shall be 5.0 V ± 5% (USB-compatible).
6. At 60 RPM, the device shall supply ≥ 400 mA continuous at 5 V (≈ 2 W).
7. The device shall sustain ≥ 2 minutes of continuous charging at 60–90 RPM without thermal shutdown.
8. The device shall handle load steps 0–1 A with output not below 4.75 V for more than 50 ms.
9. The device shall include over-voltage, over-current, and reverse-polarity protection.
10. In an emergency, 10 minutes of cranking at 60–90 RPM shall deliver ≥ 600 mWh to a typical smartphone.

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User Feedback (Display)

11. The device shall display output voltage with 0.1 V resolution and ±0.2 V accuracy.
12. The device shall display crank RPM with 1 RPM resolution and ±5 RPM accuracy (20–120 RPM).
13. The display shall update ≥ 4 times per second and be readable outdoors.

Ergonomics & Operation

14. Required continuous hand torque at 60 RPM shall be ≤ 0.35 N·m.
15. The crank shall be usable by right- or left-handed users without reconfiguration.
16. Setup time (place/hold, plug in, start cranking) shall be ≤ 20 seconds.
17. The base shall be non-slip for tabletop use; the housing shall provide a secure grip for in-hand use.
18. The crank mechanism shall help users maintain 40–90 RPM (e.g., smooth feel or light detents).

Interfaces & Compatibility

19. The device shall provide a USB-A output port and support standard phone cables.
20. No sharp edges or pinch points shall be exposed during use.

Durability

21. The device shall withstand 50 N side load (pocket compression) without damage.
22. The device shall survive 3 drops from 1.0 m (any face/edge/corner onto plywood over concrete) with full function retained.

Environmental

23. The device shall operate from 40–100 °F and 15–85% RH (non-condensing).
24. The enclosure shall resist light dust/debris and be wipe-clean without disassembly.

Safety

25. User-touch surfaces shall stay ≤ 48 °C (118 °F) during continuous operation at 90 RPM in 85 °F ambient.
26. Output current shall be limited to ≤ 1.5 A at 5 V.
27. Skin-contact surfaces shall be smooth (no burrs; finish ≈ Ra ≤ 6.3 μm).

Reliability & Maintenance

28. ≥ 95% of charging sessions shall reach and maintain a stable 5 V.
29. The crank and gears shall have MTTF ≥ 100,000 revolutions.
30. Voltage/RPM displays shall be available ≥ 99% of the time during a session.
31. The output cable shall be user-replaceable (standard cable).
32. Exterior cleaning with a damp cloth + mild detergent shall not cause damage.
33. Basic service (open, replace part, close) shall take ≤ 10 minutes using basic tools.

Verification

34. Each requirement shall have a defined verification method (Test, Analysis, Inspection, or Demonstration) and pass/fail criteria.
35. Scenario tests shall confirm:

- Tabletop use (stable, non-slip, meets power spec),
- In-hand use (comfortable torque, size/weight, pocket compression),
- Post-drop emergency use (still charges, displays are readable, and safety limits are met).

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5. Functional Description

For the Human-Powered Charger (HPC) to perform as intended, it must interact with the external entities outlined in the Context Diagram: Device, Charging Cord, User, and Environment. These entities represent all approved points of interaction. The HPC must also prevent unsafe or unintended interactions, such as overvoltage, overheating, or unsafe mechanical use.

Interactions with these entities are enabled through the following top-level functions. All functions of the HPC fall under one or more of these categories:

- Receive mechanical input
- Convert mechanical energy to electrical energy
- Regulate electrical output
- Transfer electrical energy to the connected Device.
- Monitor and display system information such as power output and RPM
- Detects and manages circuit load to ensure safe and efficient energy transfer.
- Store, distribute, or buffer power
- Protect against environmental stress
- Facilitate safe physical connection
- Prevent unsafe operation

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5.1 Requirements-Functions Trace Matrix

		Top Level Functions										
HPC Operational Requirements		Receive mechanical input	Convert mechanical energy to electrical energy	Regulate electrical output	Transfer electrical energy to the connected Device	Monitor and display system information such as power output and RPM	manage circuit load to ensure safe and efficient energy transfer	Store, distribute, or buffer power	Protect against environmental stress	Facilitate safe physical connection	Prevent unsafe operation	Provide a comfortable experience for the user
General												
1	The charger shall work using only hand cranking (no wall power required)	**X**										
2	The charger shall start delivering power within 5 seconds of cranking	**X**	**X**	**X**			**X**	**X**				
Size & Weight												
3	The device shall fit within 6.0 × 3.0 × 1.5 in when stored											**X**
4	The device shall weigh ≤ 350 g (0.77 lb)											**X**
Power & Electrical												
5	Output shall be 5.0 V ± 5% (USB-compatible)			**X**	**X**			**X**				
6	At 60 RPM, the device shall supply ≥ 400 mA continuous at 5 V (≈ 2 W)		**X**	**X**	**X**		**X**	**X**				
7	The device shall sustain ≥ 2 minutes of continuous charging at 60–90 RPM without thermal shutdown.				**X**		**X**	**X**			**X**	

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8	The device shall handle load steps 0–1 A with output not below 4.75 V for more than 50 ms.						**X**				**X**	
9	The device shall include over-voltage, over-current, and reverse-polarity protection.											
10	In an emergency, 10 minutes of cranking at 60–90 RPM shall deliver ≥ 600 mWh to a typical smartphone.	**X**	**X**	**X**	**X**			**X**				
User Feedback (Display)												
11	The device shall display output voltage with 01 V resolution and ±0.2 V accuracy.					**X**						**X**
12	The device shall display crank RPM with 1 RPM resolution and ±5 RPM accuracy (20–120 RPM)					**X**						**X**
13	The display shall update ≥ 4 times per second and be readable outdoors					**X**						**X**
Ergonomics & Operation												
14	Required continuous hand torque at 60 RPM shall be ≤ 0.35 N·m	**X**										**X**
15	The crank shall be usable by right- or left-handed users without reconfiguration.	**X**										**X**
16	Setup time (place/hold, plug in, start cranking) shall be ≤ 20 seconds	**X**								**X**		**X**
17	The base shall be non-slip for tabletop use; the housing shall provide a secure grip for in-hand use.							**X**	**X**			**X**
18	The crank mechanism shall help users maintain 40–90 RPM (e.g., smooth feel or light detents)	**X**										**X**

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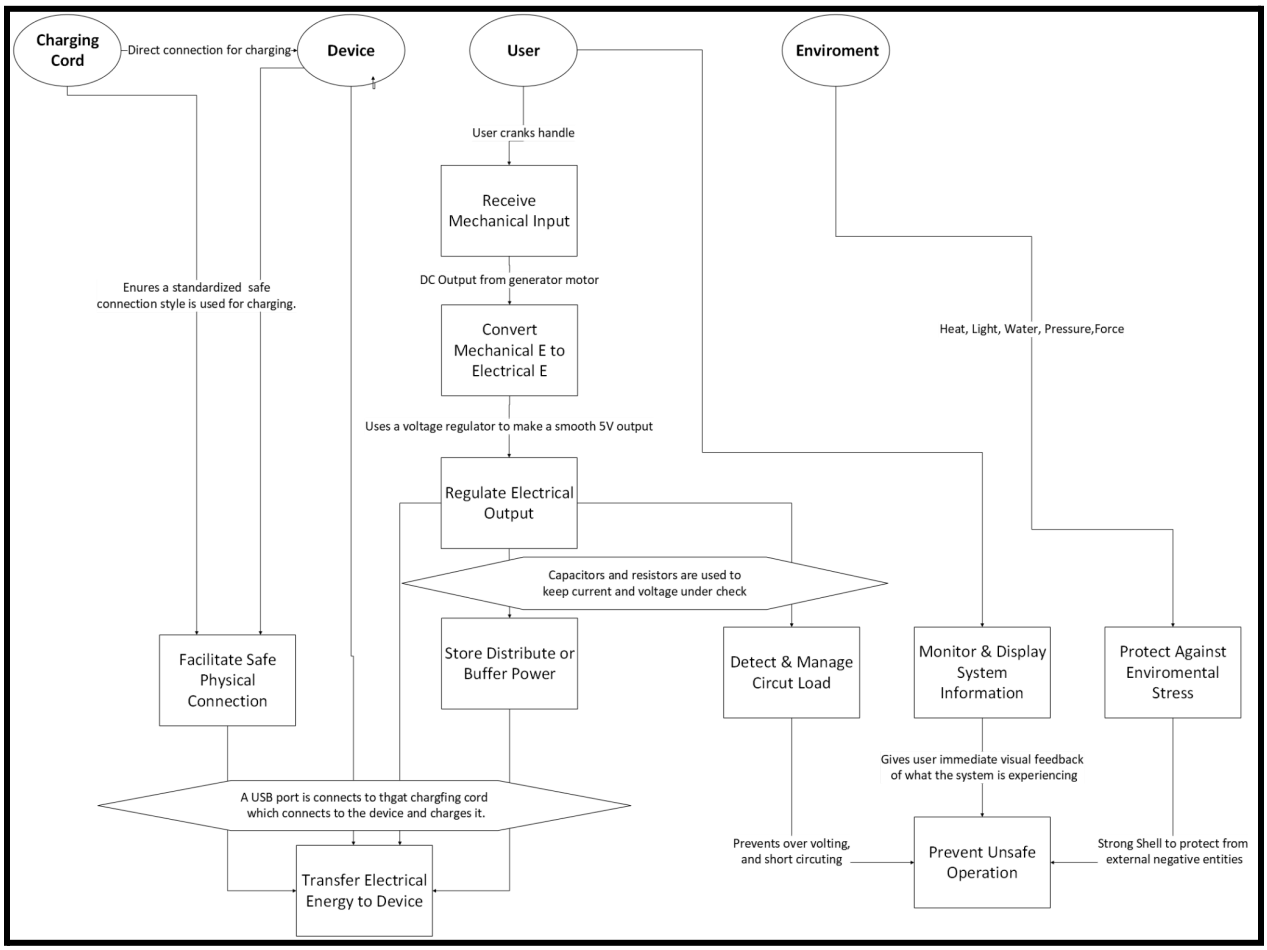
Interfaces & Compatibility											
19	The device shall provide a USB-A output port and support standard phone cables.				**X**					**X**		**X**
20	No sharp edges or pinch points shall be exposed during use										**X**	**X**
Durability											
21	The device shall withstand 50 N side load (pocket compression) without damage.								**X**		**X**	**X**
22	The device shall survive 3 drops from 1.0 m (any face/edge/corner onto plywood over concrete) with full function retained.								**X**		**X**	
Environmental											
23	The device shall operate from 40–100 °F and 15–85% RH (non-condensing)								**X**	**X**		
24	The enclosure shall resist light dust/debris and be wipe-clean without disassembly.								**X**			
Safety											
25	User-touch surfaces shall stay ≤ 48 °C (118 °F) during continuous operation at 90 RPM in 85 °F ambient								**X**	**X**		**X**
26	Output current shall be limited to ≤ 1.5 A at 5 V			**X**	**X**		**X**	**X**		**X**		
27	Skin-contact surfaces shall be smooth (no burrs; finish ≈ Ra ≤ 6.3 µm)								**X**	**X**		**X**
Reliability & Maintenance											

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28	≥ 95% of charging sessions shall reach and maintain a stable 5 V			**X**	**X**		**X**	**X**				
29	The crank and gears shall have MTTF ≥ 100,000 revolutions	**X**							**X**			**X**
30	Voltage/RPM displays shall be available ≥ 99% of the time during a session					**X**						**X**
31	The output cable shall be user-replaceable (standard cable)									**X**		**X**
32	Exterior cleaning with a damp cloth + mild detergent shall not cause damage.								**X**			**X**

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5.2 Functional Block Diagram



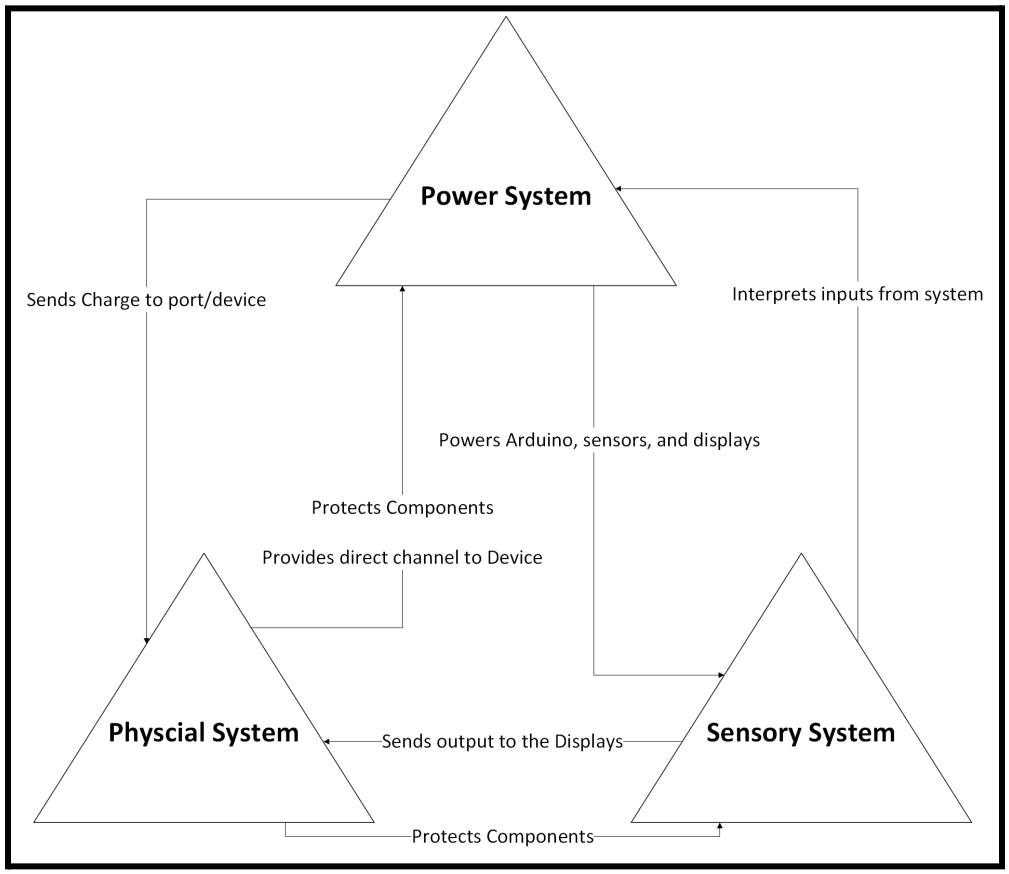
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6. Physical Description

The physical description of the HPC is aided by the system-level block diagram of Figure 6.1 and the subsystem and component level in Figure 6.2

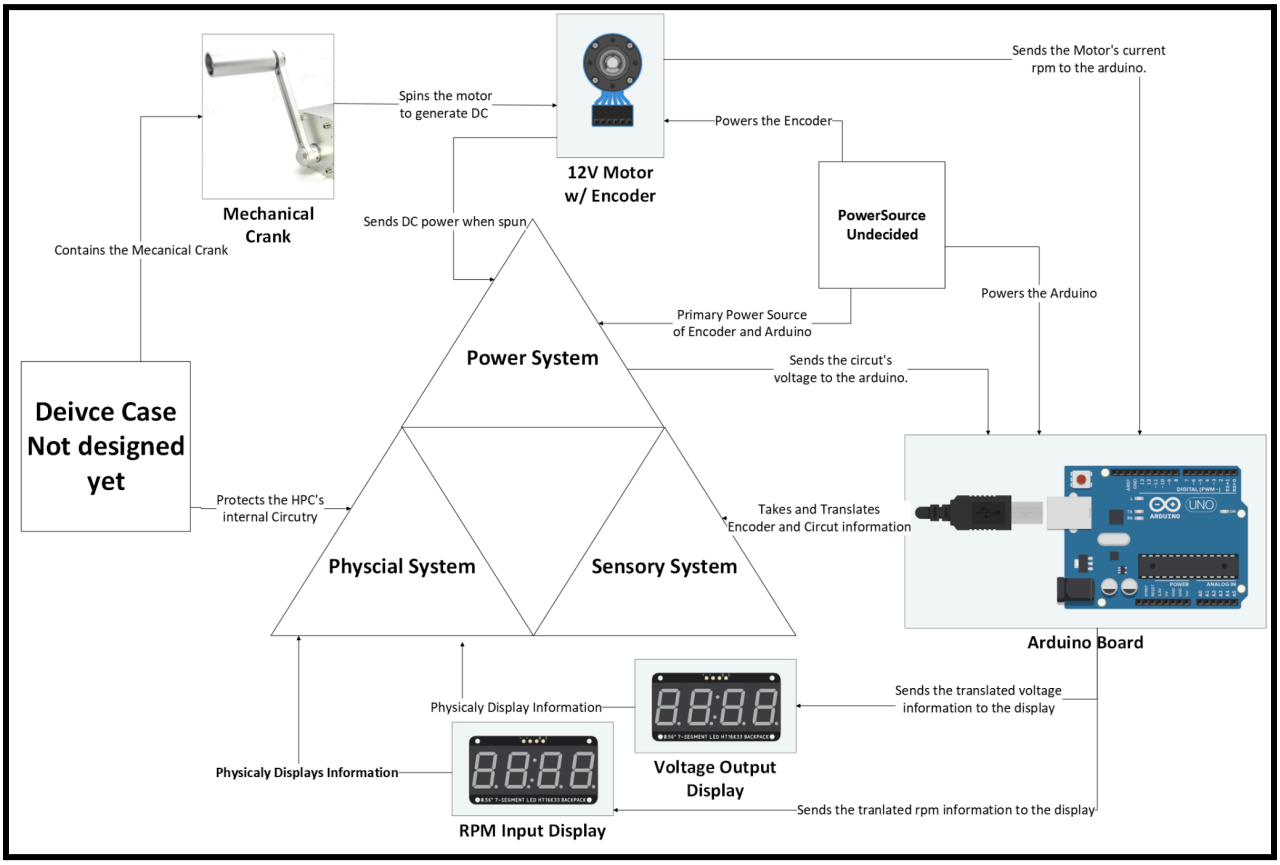
6.1 System Level Physical Description

There are 3 sub-systems in the HPC: the Power System, the Sensory System, and the Physical System. The Power System is the movement of power through the HPC, this being the power to the charger and the power to the Arduino and encoder. The Sensory System includes the inputs and outputs of the HPC. Centralised by the Arduino, the rpm and voltage from the crank and motor are sent to the Arduino to be outputted by the Arduino. Displays. The physical system includes the components that directly interact with the user and exterior hazards. Centralised by the case, the physical system holds together the entire HPC, contains the crank used to generate power, and holds the displays shown to the user.



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6.2 Physical description of the sub-system and component level



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6.3 Requirement Allocation Tables

Req #	Requirement	Physical System	Sensory System	Power System
1	The charger shall work using only hand cranking (no wall power required)	**X**		**X**
2	The charger shall start delivering power within 5 seconds of cranking.		**X**	**X**
3	The device shall fit within 6 × 3 × 1.5 in when stored.	**X**		
4	The device shall weigh ≤ 350 g (0.77 lb).	**X**		
5	Output shall be 5.0 V ± 5% (USB-compatible).		**X**	**X**
6	At 60 RPM, the device shall supply ≥ 400 mA continuous at 5 V (≈ 2 W).		**X**	**X**
7	The device shall sustain ≥ 2 minutes of continuous charging at 60–90 RPM without thermal shutdown.	**X**	**X**	**X**
8	The device shall handle load steps 0–1 A with output not below 4.75 V for more than 50 ms.		**X**	**X**
9	The device shall include over-voltage, over-current, and reverse-polarity protection.		**X**	**X**
10	In an emergency, 10 minutes of cranking at 60–90 RPM shall deliver ≥ 600 mWh to a typical smartphone.	**X**		**X**
11	The device shall display output voltage with 0.1 V resolution and ±0.2 V accuracy.	**X**	**X**	
12	The device shall display crank RPM with 1 RPM resolution and ±5 RPM accuracy (20–120 RPM).	**X**	**X**	
13	The display shall update ≥ 4 times per second and be readable outdoors.	**X**	**X**	**X**

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Req #	Requirement	Physical System	Sensory System	Power System
14	Required continuous hand torque at 60 RPM shall be ≤ 0.35 N·m.	**X**		
15	The crank shall be usable by right- or left-handed users without reconfiguration.	**X**		
16	Setup time (place/hold, plug in, start cranking) shall be ≤ 20 seconds.	**X**		
17	The base shall be non-slip for tabletop use; the housing shall provide a secure grip for in-hand use.	**X**		
18	The crank mechanism shall help users maintain 40–90 RPM (e.g., smooth feel or light detents).	**X**	**X**	
19	The device shall provide a USB-A output port and support standard phone cables.	**X**		**X**
20	No sharp edges or pinch points shall be exposed during use.	**X**		
21	The device shall withstand 50 N side load (pocket compression) without damage.	**X**		
22	The device shall survive 3 drops from 1.0 m (any face/edge/corner onto plywood over concrete) with full function retained.	**X**		
23	The device shall operate from 40–100 °F and 15–85% RH (non-condensing).	**X**	**X**	**X**
24	The enclosure shall resist light dust/debris and be wipe-clean without disassembly.	**X**		
25	User-touch surfaces shall stay ≤ 48 °C (118 °F) during continuous operation at 90 RPM in 85 °F ambient.	**X**	**X**	**X**
26	Output current shall be limited to ≤ 1.5 A at 5 V.		**X**	**X**
27	Skin-contact surfaces shall be smooth (no burrs; finish ≈ Ra ≤ 6.3 µm).	**X**		
28	≥ 95% of charging sessions shall reach and maintain a stable 5 V.		**X**	**X**

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Req #	Requirement	Physical System	Sensory System	Power System
29	The crank and gears shall have MTTF ≥ 100,000 revolutions.	**X**		
30	Voltage/RPM displays shall be available ≥ 99% of the time during a session.	**X**	**X**	**X**
31	The output cable shall be user-replaceable (standard cable).	**X**		
32	Exterior cleaning with a damp cloth + mild detergent shall not cause damage.	**X**		
33	Basic service (open, replace part, close) shall take ≤ 10 minutes using basic tools.	**X**		

	Components							
Primary Function	Mechanical Crank	HPC Enclosure	12V Motor and Encoder	Sensor power source	Arduino	Voltage Output Display	RPM Output Display	USB Port
Receive mechanical input	X		X					
Convert mechanical energy to electrical energy	X		X					
Regulate electrical output				X	X			
Transfer electrical energy to the connected Device.			X					X
Monitor and display system information such as power output and RPM			X	X	X	X	X	
Detects and manages circuit load to ensure safe and efficient energy transfer.				X	X			X

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Store, distribute, or buffer power				X				
Protect against environmental stress.		X		X				
Facilitate safe physical connection.		X						X
Prevent unsafe operation	X	X		X	X			X

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7. Interface Description

7.1 Power System & Sensory System

The interfaces between the Power System and the Sensory System are shown in Table 7.4.1.

The Power System is responsible for supplying regulated power both to the device connected to the USB output and to the Sensory System's sensors, encoders, and Arduino. The Sensory System uses a current sensor integrated into this circuitry to interpret current flow and calculate wattage for user feedback. It also uses the Arduino to process voltage and RPM signals originating from the Power System. The Power System depends on the Sensory System to diagnose electrical irregularities, collect power-related data, and translate that data into readable output for the user. In turn, the Sensory System relies on the Power System to provide stable and sufficient power to its sensors and microcontroller so it can accurately measure and display performance metrics. The physical interface between these two systems consists of electrical wiring, signal routing, and a shared ground connection.

7.2 Power System & Physical System

The interfaces between the Power System and the Physical System are shown in Table 7.4.1.

The Physical System houses the mechanical components—such as the crank, shaft, bearings, gear train, and enclosure—that allow the generator to be driven at the required RPM range. These mechanical elements directly enable the Power System to convert mechanical rotation into electrical power. The Physical System also supports and protects the USB port, voltage regulation circuitry, and wiring pathways that carry power from the generator to the output device. The Power System relies on the Physical System to maintain proper generator alignment, structural rigidity, and ergonomic access to the crank. Without these physical supports, consistent voltage and current output would not be achievable. Conversely, the Physical System relies on the Power System to provide safe, regulated electrical output that does not exceed thermal limits of enclosure materials or the USB components. The interface between the two systems consists of mechanical mounting points, shaft couplings, port alignment features, and secure cable routing.

7.3 Sensory System & Physical System

The interfaces between the Sensory System and the Physical System are outlined in Table 7.4.1.

The Physical System provides the supporting structure and mounting points for the Arduino, current sensor, encoder, and display module. It ensures that the RPM sensor is correctly aligned with the crankshaft and that the display is visible and accessible to the user during operation. Cable routing channels and enclosure cutouts also originate from the Physical System, aiding in protecting and guiding the Sensory System's wiring. The Sensory System relies on the Physical System to maintain stable sensor alignment, prevent vibration-induced measurement errors, and ensure durability under drops, side loads, and varying environmental conditions. In return, the Sensory System provides real-time feedback—such as voltage, current, wattage, and RPM—which allows users to understand device performance and identify when mechanical or electrical issues may occur. The physical interface between these systems consists of sensor mounts, display windows, wiring channels, and enclosure features that secure and protect the Sensory System hardware.

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Table 7.4.1 System Interface Description

System Interface Description			
Subsystem	**Interface**	**Data transferred from Subsystem to Interface**	**Data transferred from the Interface to the Subsystem**
Power System	Voltage Regulator	Unregulated generator voltage	Regulated 5V output feedback, load response behaviour
	Current Sensor (electrical inline interface)	Raw current flow through the load line	Current measurement signal for data interpretation
	DC Motor Output Leads	DC Voltage and Current	Electrical load feedback (voltage drop under load)
	USB Output Port	Regulated 5 V, up to 1.5 A	Physical connection status, load detection
	Shared Ground Plane	Reference ground for all signals	Stabilised ground return for regulators, sensors, Arduino
Sensory System	Arduino Microcontroller	Processed voltage/current/RPM readings, display outputs	Regulated 5V input power from the Power System
	Current Sensor	Current interpretation signal (voltage proportional)	Raw current flow routed through the sensor shunt
	RPM Encoder	Pulse output for rotational speed data	Mechanical rotation (from crank via Physical System)
	Display Module	Visual wattage and RPM output	Power and signal from Arduino
Physical System	DC Motor & Crank	Mechanical rotation, torque transfer	User input force, rotational resistance feedback
	Enclosure	Mounting structure, environmental protection	Heat load from electronics, vibration from the crank
	Display Window	Physical protection and visibility path	Display brightness & user visibility constraints
	Wiring Channels	Physical routing for cables	Cable stiffness, thermal expansion, vibration transfer

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Table 7.5.1 System Implementation

Interface Implementation		
Subsystem	**Interface**	**Ways to Implement Data Transfer**
Power System	Voltage Regulator	Use a low-dropout regulator; include heat sinking; add input/output capacitors to stabilise voltage.
	Current Sensor (electrical inline interface)	Place sensor on high-side; route shielded wires; calibrate in software via Arduino.
	DC Motor Output Leads	Secure wiring to minimise vibration-related noise.
	USB Output Port	Reinforced mount; short, thick wiring to minimise voltage drop.
	Shared Ground Plane	Use a common ground bus; keep the sensor ground near the Arduino to reduce noise loops.
Sensory System	Arduino Microcontroller	Convert the analog signal to a digital output
	Current Sensor	Use the analog output pin to feed the Arduino.
	RPM Encoder	Debounce signal in software.
	Display Module	Connect to digital output pins and ensure a stable 5V output
Physical System	DC Motor & Crank	implement ergonomic handle; ensure stable connect for 80+ rpm rotations
	Enclosure	3D Printed PLA enclosure, shock-absorbent mounting posts for electronics
	Display Window	Transparent acrylic; angled for outdoor visibility; sized to match
	Wiring Channels	Provide strain relief; isolate wires from the moving crank; include clips/guides for cable routing.

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8. Handheld Crank Phone Charger (HPC) Trade Study

Today, portable energy solutions are used across a wide range of consumer, educational, outdoor, and emergency-response situations. Because of these different needs, the market for handheld or manually powered chargers includes many commercial products, each with its own strengths and limitations. Looking at these products helps guide the design of the HPC system and shows which features already exist and which still need improvement.

Most users, students, or emergency-preparedness consumers have likely seen or used some form of crank-powered charger. The fact that this project required a student-designed device shows that no single commercial option meets all the specific electrical output, durability, user feedback, size, and ergonomic requirements that were set for the HPC. Even though certain features like USB charging, manual cranking, compact size, or basic indicators are common, the full set of required capabilities is not available in one product.

A good trade study helps the team narrow down the design scope by identifying existing technologies that already solve some requirements. This reduces unnecessary development time and allows the project to focus on the areas that matter most, such as improving electrical efficiency, raising durability, adding meaningful user feedback, and creating a more ergonomic and reliable design.

8.1 Trade-Off Analysis Objectives

Trade studies are helpful throughout the entire system development process, and the purpose of the study usually depends on the current stage. During Integration and Test, the team may look at testing options or sensor calibration strategies. Later in development, the focus might shift toward fine-tuning subsystem layout, improving voltage regulation, or refining the user interface.

For the HPC, this trade study is being done during the Concept Exploration phase, which is where high-level decisions are made, and major requirements are balanced for the first time. In this phase, the goal is no longer to compare full commercial crank chargers. Instead, the focus shifts to comparing individual components and technologies, which matches the work done in Sections 8.2 through 8.5. This allows us to identify which materials, generator types, port options, display styles, and regulators will best help us meet the HPC specifications.

The objectives for this trade study are as follows:

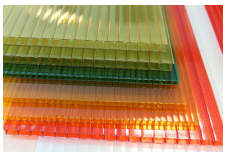
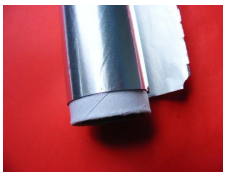
- **Identify component-level solutions that support efficient hand-crank energy conversion, such as motor type, generator structure, gear ratios, and torque requirements.**
- **Review existing regulation strategies, such as USB-A, USB-C, and Quick Charge behaviour, along with voltage stability, peak current needs, and overcurrent protection.**
- **Evaluate feedback and display options, including LEDs, LCDs, or E-Ink, to determine the best way to provide clear electrical output data without draining power.**
- **Compare enclosure materials and crank mechanics to guide decisions about durability, weight, comfort, and stowage.**
- **Consider ease of use, environmental performance, and thermal limits to ensure the HPC stays functional in the required temperature and humidity ranges.**
- **Determine which limitations seen in commercial chargers justify a custom design, including low power output, no real-time feedback, poor ergonomics, or low survivability under load or drop conditions.**

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These objectives help ensure the HPC design is not just another version of what already exists. Instead, it becomes a focused, requirement-driven system that uses the best ideas found in commercial products while directly addressing the major shortcomings they commonly have.

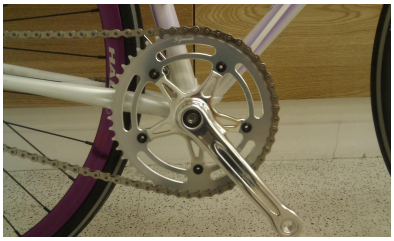
8.2 HPC Alternatives

- **Enclosure Material**: The enclosure must protect the device while being lightweight.
    - Acrylonitrile butadiene styrene (ABS) is inexpensive and easy to injection-mould, making it a cost-effective choice, but it has lower impact and temperature resistance.
    - Polycarbonate (PC) is more durable and highly impact-resistant, tolerating higher temperatures and chemicals.
    - Aluminium offers a very high strength-to-weight ratio, good thermal conduction, and corrosion resistance, but it can scratch or dent on impact and adds weight and machining cost.
    - A rubberised-plastic (thermoplastic overmold) enclosure provides excellent grip and drop cushioning (thermoplastic elastomer coatings are often used to improve ergonomics), at the expense of added manufacturing steps and cost.
    - Polycarbonate with a moulded rubbery overcoat (TPE/TPU) is a common compromise that yields strong impact resistance plus shock absorption.
- **We will evaluate durability and ruggedness for each material.**

ABS	PC	Aluminum	Rubberised- Plastic	PC + TPU
				

- **Crank Styl**e: The crank mechanism must be comfortable to use and stowable.
    - A fixed crank is simple and very robust, but a long fixed handle can make the unit bulky and hard to pack. A foldable crank (hinged or swivelling) allows the handle to collapse flat against the enclosure, improving portability; this design is common in hand tools because it balances strength with compactness.
    - A retractable (telescoping or slidable) crank can save space but adds mechanical complexity and potential failure points.
    - Foldable handles are typically recommended for consumer handheld tools because ergonomic handle shape and non-slip textures (e.g. rubber grips) improve comfort and reduce user fatigue.
- **Our study will consider all three styles, weighing the trade-off between ruggedness and packability**.

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Fixed crank	Retractable crank	Foldable crank
		

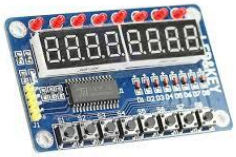
- **USB Output Standard:** The output port must match modern phone charging standards.
    - USB-A is widely compatible with existing cables, but by default is limited to 5 V/0.5 A (2.5 W) per the original USB spec. Newer Quick-Charge (QC) schemes on USB-A can negotiate higher voltages (e.g. 9 V or 12 V at 1–2 A) to charge some phones faster, but QC support is only on compatible phones.
    - USB-C is now the emerging standard: it is reversible and can support much higher power (up to 20 V at 5 A, 100 W) when paired with USB Power Delivery. For an HPC, USB-C is attractive for future compatibility and higher current (most modern phones use USB-C), while a USB-A output (with QC) could be a low-cost fallback for older devices.
- **We will evaluate providing a single USB-C port (with PD support if possible) versus a USB-A port (with or without QC) in terms of compatibility and cost.**

USB-A	USB-C
	

- Display Type: The display should inform the user (charge status, instructions) while using minimal power.
    - Traditional LCD or OLED screens offer full graphics and colour but consume continuous backlight power (LCD) or emit light (OLED), and may wash out in bright sun. LED indicators (e.g. seven-segment or dot-matrix) are simple and visible in the dark but draw current and have limited information content.
    - In contrast, reflective “memory” LCD or E-Ink displays consume almost no power when showing a static image and have excellent outdoor readability. For example, E-Ink displays use ambient light (like paper) and require energy only to change the screen, making them ideal for low-power status displays.

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- **We will compare a basic LED/LCD numeric indicator versus a simple monochrome LCD versus an E-Ink panel, evaluating factors like brightness in sunlight and power draw.**

LED	E-Ink
	

- **Generator Motor Type**: The internal generator converts crank motion to electrical power.
    - A simple brushed DC motor (acting as a generator) is cheap and easy to drive but has lower efficiency (brush friction) and shorter life (brush wear.
    - A brushless DC (BLDC) motor requires electronic commutation but offers significantly higher efficiency and longer service life.
    - Stepper motors can generate electricity but are generally bulky, noisy, and inefficient in this use (they are optimised for precise motion, not power generation).
    - A dynamo/alternator (field-coil generator) might handle higher power, but adds weight and complexity. Given the high priority on maximising charge per crank, efficiency is critical.
- **The study will compare a low-cost brushed motor, a mini BLDC (outrunner style), and possibly a small automotive alternator design, measuring open-circuit voltage and loaded output.**

Brushed DC	Brushless DC	STepper	Dynamo
			

- Voltage Regulation Type: The charger must regulate the raw generator voltage (which can vary widely with crank speed) down to a steady 5 V (± tolerance) output.
    - A linear LDO regulator is the simplest (low part count, quiet), but any voltage drop is dissipated as heat, making it very inefficient when the input is higher than 5 V.
    - A switching step-down (buck) converter can achieve much higher efficiency (it switches fully on/off and wastes very little power), but only supports output < input.
    - A buck-boost switching converter can efficiently regulate whether the generator output is above or below 5 V by switching modes, at the cost of more complex control circuitry.
- **We will evaluate simple LDO versus buck versus buck-boost converters, considering their efficiency, size, and component count for the HPC's expected power range.**

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Linear LDO	Buck Converter	Buck-Boost Converter
		

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8.3 Study Criteria

- Durability/Impact Resistance:
    - This measures how well the device survives drops, shocks, or rough handling.
    - We will consider material toughness and enclosure strength in this criterion.
        - Impact resistance is defined as the ability to withstand sudden high-force impacts without cracking or deforming.
    - In practice, we might use a drop-height test or drop energy threshold as a metric. Enclosure options will be scored by their known toughness (e.g. polycarbonate and aluminium are known for high impact strength).

- Power Efficiency:
    - This criterion measures the electrical output achieved per unit of human input (or per unit of mechanical work).
    - It includes the efficiency of the generator and regulator.
        - For example, brushless generators and switching regulators can be >80–90% efficient, whereas brushed motors or LDOs may be much lower.
    - We will calculate efficiency metrics (output power divided by input crank power) for each option. Higher efficiency scores higher utility.

- Ergonomics (Crank Comfort & Stowage):
    - Ergonomics covers user comfort in cranking and ease of packing the device.
    - Factors include handle shape, size, grip material, and the overall cranking torque. For instance, contoured, non-slip handles reduce hand strain.
        - A bulkier fixed crank may score lower on stowage, whereas a foldable handle scores higher.
    - We will rate designs on grasp comfort and compactness (e.g. volume with crank stowed). Qualitative data (e.g. user ratings or anthropometric fit) may be used.

- Display Readability & Power Use:
    - We will measure how easily information is read on the display under various lighting conditions, and how much power the display consumes.
        - For example, E-Ink has excellent sunlight readability and near-zero idle current, while LCD/OLED screens have continual power draw for backlighting.
    - Criteria may include contrast ratio in bright light and current draw at typical refresh rates. Displays that use less power and remain legible in sunlight will score higher.

- Port Compatibility (Modern Phones & Cables):
    - This user-centric criterion evaluates how well the output port matches target devices.
        - A USB-C port inherently supports the newest phones and can handle higher voltages, giving it a high compatibility score.
        - USB-A (with QC) may not fit all phones natively, but supports legacy devices.
    - We will consider the fraction of current smartphone models that can charge directly (and the need for adapters). Higher compatibility scores reflect wider phone acceptance (e.g. USB-C vs USB-A).

- Manufacturability (Cost and Complexity):
    - We assess how easy and economical it is to produce each option. Simpler solutions with fewer and cheaper parts score higher.
        - For example, a plastic enclosure injection-moulded in one shot is cheaper than a machined metal box. An LDO regulator is “simple, low cost”, whereas a buck-boost converter or overmolded enclosure adds parts and steps.
    - We will estimate relative material and assembly cost, considering factors like specialised tooling or multi-stage processes (e.g. overmolding adds expense). Lower manufacturing cost and complexity yield higher scores.

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8.3.5 Study criteria Data Sources

Sources

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8.4 Study Criteria Weighting

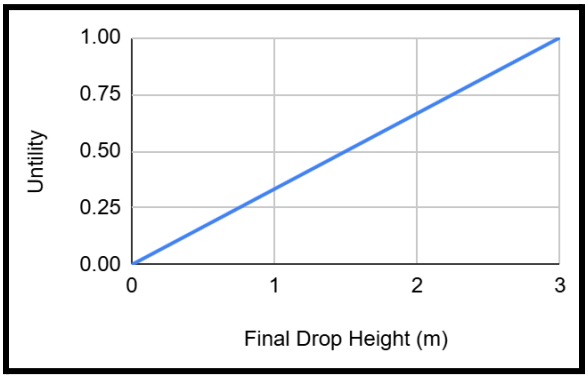
8.4.1 Weighing Style

We assign weights on a 1–10 scale to reflect the relative importance of each criterion (total importance normalised in analysis). Durability and Power Efficiency are highest (weight 9) since the device must reliably deliver energy under rugged use. Port Compatibility is also high (8) to ensure the charger works with modern phones. Ergonomics is important (weight 7) to keep the device user-friendly during hand-cranking. Display Readability is moderate (6) because while useful, it is secondary to the core function. Manufacturability is lower (5) but still considered for cost control.

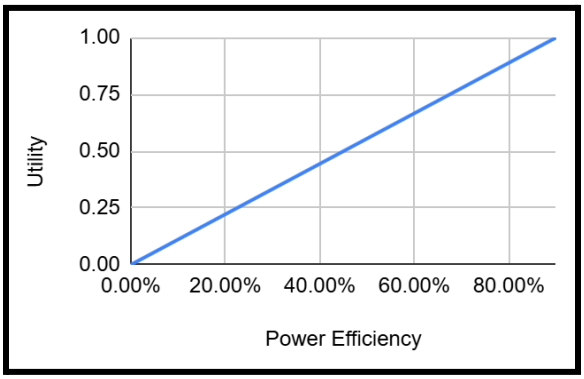
- Durability - 9
- Power Efficiency - 9
- Port Compatibility - 8
- Ergonomics - 7
- Display Readability - 7
- Manufacturability - 5

8.4.2 Utility Function

For each criterion, we define a utility function to convert a metric into a normalised score (0–1). Utility functions are a method by which one may combine criteria with different performance measurement units into a single score, such that all criteria may be evaluated as a whole. Utility functions will generate a utility score based on the minimum and maximum performance value of the given criteria in comparison with the specifications of each design. The utility functions utilised for each criterion are shown below.

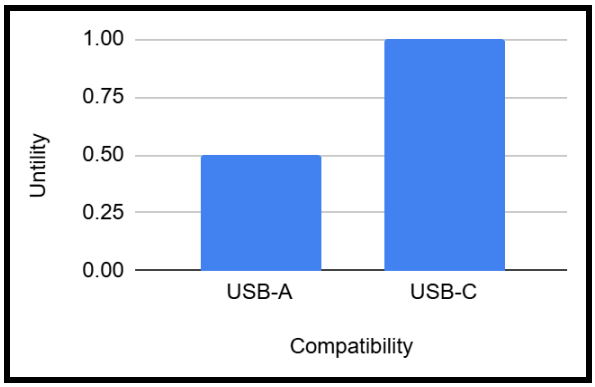


Durability Utility Function (Linear)

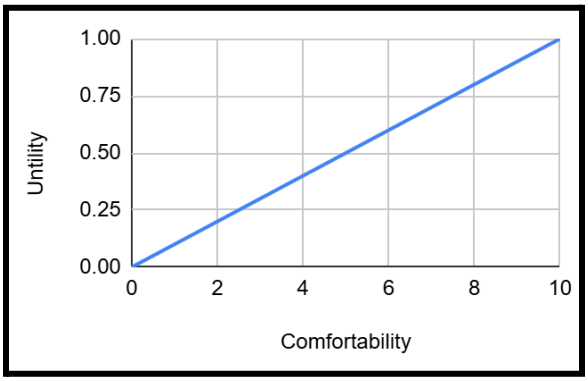


Power Efficiency Utility Function (Linear)

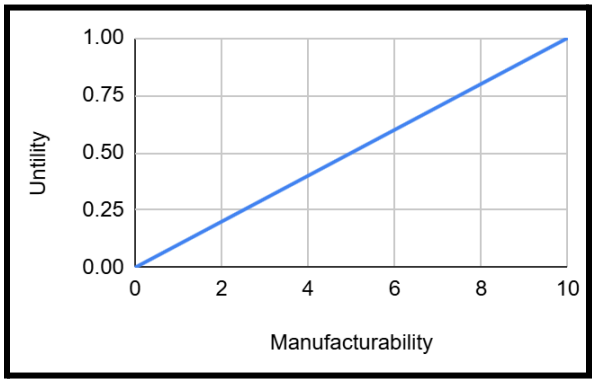
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Compatibility Utility Function (Binary)



Ergonomics Utility Function (Linear)



Manufacturability Utility Function (Linear)

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8.5 Trade Study Design Selection

Applying the weighted scoring, the most attractive choices emerged as follows:

- Enclosure Material: Polycarbonate (with a rubberised overmold) scored best. Its high impact toughness and heat tolerance (superior to ABS) aligned with our high durability weight, while the rubber coating improved grip and shock absorption. Aluminium, although strong, was penalised for weight and fabrication cost. ABS scored lower due to its lower environmental resistance. Thus, the PC/TPE combination offers the best balance of protection, weight, and cost.

- Crank Style: A foldable crank was chosen. This style allows compact stowage (meeting ergonomics) while maintaining a rigid, non-sloppy crank when deployed. Fixed cranks were less favourable because they make the unit bulky, and retractable (telescoping) handles introduced too much mechanical complexity. The foldable design meets the system's needs with only a minor durability risk at the hinge.

- USB Output: USB-C was selected as the primary output. It inherently supports high power delivery and is the current smartphone standard, maximising compatibility. Although USB-C circuitry is slightly more complex, its future-proofing and reversible connector were deemed worth the trade-off. We plan to support legacy devices via standard USB-C to Micro-USB/Lightning adapters if needed. A dedicated QC-enabled USB-A port was not chosen due to its lower universality.

- Display Type: A reflective low-power display (such as E-Ink or memory LCD) was favoured. E-Ink in particular has essentially zero static power draw and "excellent" sunlight readability, matching our emphasis on battery life and outdoor use. This choice meant sacrificing full-colour graphics, but the simple monochrome display meets user needs for status info. Conventional LCD/OLED types were rejected for their higher power draw; a simple LED indicator option was still considered as a backup due to minimal electronics.

- Generator Motor Type: A brushless DC (BLDC) motor won out. BLDC motors offer high efficiency and long life, which directly improves our energy per crank objective. The study showed that a BLDC generator produces significantly more current at a given RPM than an equivalent brushed motor. Though the BLDC requires control electronics (raising complexity and cost), the efficiency gain (on the order of 20–30% better than brushed) was worth it. Stepper motors and small alternators were inferior due to size or inefficiency.

- Voltage Regulation Type: A buck-boost switching regulator was selected. This configuration achieves very high electrical efficiency (switching losses are small) and can maintain a 5 V output even when the crank generator voltage dips below or above the target. In our analysis, the buck-boost's efficiency advantage outweighed its higher part count. An LDO, while simple and low-noise, would have wasted too much power as heat (since generator voltage can be much higher than 5 V). The buck-boost approach delivers the required voltage regulation without sacrificing our stringent efficiency requirements.

In summary, the trade study recommends the following design: a polycarbonate enclosure with rubber overmold, a foldable crank handle, a USB-C output port, a reflective low-power display (E-Ink), a brushless DC generator, and a buck-boost DC–DC converter. These components collectively best satisfy the weighted system objectives. The main trade-offs involved increased complexity or cost for higher performance: for example, the BLDC generator and switching regulator add electronics compared to a simple motor+LDO, but they greatly improve power output efficiency. Likewise, the premium enclosure materials meet durability goals at a modest cost increase. Overall, the selected options align well with the end-user priorities (durability, energy efficiency, usability) as determined by our weighted criteria.

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9. Risk Management Plan

9.1 Likelihood & Consequences

Likelihood		
Level	Probability	Approach
1	Low	Will avoid or mitigate based on standard practices.
2	Low-Medium	Have mitigated this type of risk with minimal oversight previously
3	Medium	May mitigate the risk, workaround required
4	Medium-High	Cannot mitigate this risk, but a different approach can
5	High	Cannot mitigate this risk; no workaround available

Consequences				
Level	Technical Performance	System Project Schedule	Cost	Impact on Other
Low	Minimal or no impact	<2 weeks' impact on the schedule	<1% of Maintenance Reserve	Minimal
Low-Medium	Minor Impact	<1 month minor delays	1-5% of Maintenance Reserve	Minor Impact
Medium	Moderate impact with workarounds	1 month delay; missed deadlines	5-20% of Maintenance reserve	Moderate Impact
Medium- High	Unacceptable impact with workarounds available	> 1month delay; project schedule affected	Maintenance Reserve	Major Impact
High	Unacceptable with no alternatives	Unable to meet key project milestones	Budget Increase	Unacceptable Impact

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9.2 Risk Assesment Charts

Production Risks (Pre - Pre-Mitigation)				
Item	Risk	Likelihood	Consequence	Mitigation Strategy
P1	Generator (motor) output varies due to inconsistent manufacturing tolerances	Medium	Low	Source motors from a reliable supplier, perform batch-level QC testing
P2	PCB or solder joint failure	Medium	High	Automated reflow inspection, manual QC on sensitive joints
P3	Enclosure deformation or cracking during assembly	Low–Medium	Medium	Use PC/TPE materials, enforce proper torque specs during assembly
P4	The crank hinge or pivot mechanism fails early	Medium	Medium–High	Increase hinge thickness, use reinforced pin, cycle-test hinges
P5	Voltage regulator overheating during continuous use	Medium	High	Heat spreading copper, a higher-efficiency buck-boost converter, and airflow gaps
P6	Display misalignment or damage during installation	Low	Medium	Add alignment posts, install the display after the major mechanical assembly
P7	Wiring harness routing errors are causing shorts	Low–Medium	High	Colour-coded wiring, routing channels, and final continuity checks
P8	USB-C port solder pads lifting or breaking during assembly	Low–Medium	High	Reinforced port frame, through-hole mechanical supports

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Consumer Risks (Pre- Pre-Mitigation)				
Item	Risk	Likelihood	Consequence	Mitigation Strategy
C1	User over-cranks, causing motor overspeed	Medium	Medium	Mechanical gearing limit, regulator over-voltage protection
C2	The device slips from the user's hand during use	Medium	Low–Medium	Rubberised TPE overmold, ergonomic 3-contoured grip
C3	Finger pinch hazard at crank hinge	Low–Medium	Medium	Protective hinge covers, pinch-safe geometry
C4	Device fails after accidental drop	High	Medium–High	PC/TPE enclosure, internal shock absorption ribs
C5	Overheating during prolonged cranking	Medium	Medium–High	Venting gaps, thermal regulation on the buck-boost converter
C6	Display becomes unreadable in outdoor/sunlight conditions	Medium	Low	E-Ink display selection, anti-reflective cover
C7	USB-C cable incorrectly inserted or stressed	Medium	Medium	Reinforced port, internal cable strain relief
C8	User receives inaccurate power feedback (voltage, current, RPM)	Low–Medium	Medium	Calibrated sensors, software averaging, redundant readings

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Production Risks (Post-Mitigation)			
Item	Risk	Likelihood (After)	Consequence (After)
P1	Generator (motor) output varies due to inconsistent manufacturing tolerances	Low	Low
P2	PCB or solder joint failure	Low	Medium
P3	Enclosure deformation or cracking during assembly	Low	Low
P4	The crank hinge or pivot mechanism fails early	Low–Medium	Medium
P5	Voltage regulator overheating during continuous use	Low	Medium
P6	Display misalignment or damage during installation	Low	Low
P7	Wiring harness routing errors are causing shorts	Low	Medium
P8	USB-C port solder pads lifting or breaking during assembly	Low	Medium

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Consumer Risks (Post Mitigation)			
Item	Risk	Likelihood (After)	Consequence (After)
C1	User over-cranks, causing motor overspeed	Low	Low–Medium
C2	The device slips from the user's hand during use	Low	Low
C3	Finger pinch hazard at crank hinge	Low	Low–Medium
C4	Device fails after accidental drop	Medium	Medium
C5	Overheating during prolonged cranking	Low–Medium	Medium
C6	Display becomes unreadable in outdoor/sunlight conditions	Low	Low
C7	USB-C cable incorrectly inserted or stressed	Low	Low
C8	User receives inaccurate power feedback (voltage, current, RPM)	Low	Low–Medium

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9.3 Pre Mitigation Risk Heat Matrix

		Pre-Mitigation Production Risks				
Likelihood	5					
	4					
	3	1			4	2,5
	2			3		7,8
	1			6		
		Low	Low-Medium	Medium	Medium-High	High
		Consequence				

		Pre-Mitigation Consumer Risks				
Likelihood	5				4	
	4					
	3	6	2	1,7	5	
	2			3,8		
	1					
		Low	Low-Medium	Medium	Medium-High	High
		Consequence				

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9.4 Post Mitigation Risk Heat Charts

		Post-Mitigation Production Risks				
Likelihood	5					
	4					
	3					
	2			3,4		
	1	1,3,6		2,5,7,8		
		Low	Low-Medium	Medium	Medium-High	High
		Consequence				

		Post-Mitigation Consumer Risks				
Likelihood	5					
	4					
	3			4		
	2			5		
	1	2,6,7	1,3,8			
		Low	Low-Medium	Medium	Medium-High	High
		Consequence				

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10. Test and Evaluation Plan

10.1 Introduction

Section 10 provides the Test & Evaluation (T&E) plan for the Handhelp Crank Phone Charger (HPC). T&E objectives are listed in the following section. Test event are in section 10.3.5.

10.1.1 Purpose

The T&E plan ensure that HPC satufies its mission description. In order to fufill these, the following T&E objectives have been established:

1. Verify that the HPC system satisfies all conditions that are defined before each test phase.
2. Validate that the HPC system satisfies all customer, user, regulator, and investor needs.
3. Evaluate that the HPC system satisfies all behavior, performance, and efficiency guidelines in order reduce system risks.
4. Experiment with concept and performance boundary operations in order to test principles and hypothesis.
5. Demonstrate that the HPC system satisfies all behavior, performance, and efficiency guidelines and successfully interacts with all necessary entities.
6. Assess the HPC system after deployment.

10.1.2 Mission Description

The Handheld Crank Phone Charger (HPC) is designed to provide reliable, user powered energy in situations where traditional power sources are not available. The HPC supports students, outdoor users, and emergency scenarios by offering a compact and durable way to generate phone charging power on demand. It is expected to operate in a variety of real world conditions, including indoor environments, outdoor settings, and mild weather exposure. The HPC will supply stable 5 volt power to modern devices through a USB connection while giving the user live feedback through voltage, RPM, and power readings.

The charger is intended to keep a device powered until the user no longer needs it or until the unit has reached thermal or ergonomic limits. The HPC meets the customer needs defined in Section 4.1 by focusing on output stability, durability, ease of use, ergonomic cranking, and clear sensory feedback during operation.

10.1.3 System Description

The HPC is built around three main subsystems: the Power System, the Sensory System, and the Physical System. The Power System handles electrical generation and regulation. The Sensory System interprets electrical data and provides real time display feedback. The Physical System includes the enclosure, crank, mounts, and structural support. Together, these subsystems work to convert human mechanical energy into safe and stable phone charging power.

All subsystems are integrated within the handheld device itself. Unlike remote or distributed systems, the HPC operates as a single compact unit that the user fully interacts with. The external interfaces, such as user interaction points and the USB output connection, are represented in the HPC Context Diagram and Functional Block Diagram.

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10.2 Test Program Management

Several groups will take part in testing and evaluation of the HPC. These groups work together throughout each test phase to make sure the system performs the way it was designed to.

- Systems Engineer
- Design Engineers
- Engineering Specialists
- Data Analysts
- Independent Testers
- Customers and Users

Systems Engineers oversee the entire Test and Evaluation Plan. They manage schedules, organize test events, approve scenarios, and make sure the system is being tested correctly at every stage. They also coordinate how discrepancies are handled when something does not perform as expected.

Design Engineers support Integration, Development, and Operational testing. These engineers are responsible for creating the system and solving technical issues during testing. If problems appear, they either resolve them directly or identify temporary workarounds.

Engineering Specialists include: safety, reliability, and maintainability engineers. They support all testing phases and work closely with Systems Engineers and Independent Testers, especially when the focus shifts toward long term durability and safety.

Data Analysts support test planning by identifying which performance data must be collected to confirm system behavior. They help define exit criteria before testing begins and later review the collected data to help evaluate the system during every test phase.

Independent Testers participate during the Operational phase. These testers are not connected to the design or systems engineering teams, so they provide unbiased evaluation. Their job is to run the established tests on their own, interact with the human interface features, and confirm that the HPC is usable, safe, and reliable for real users.

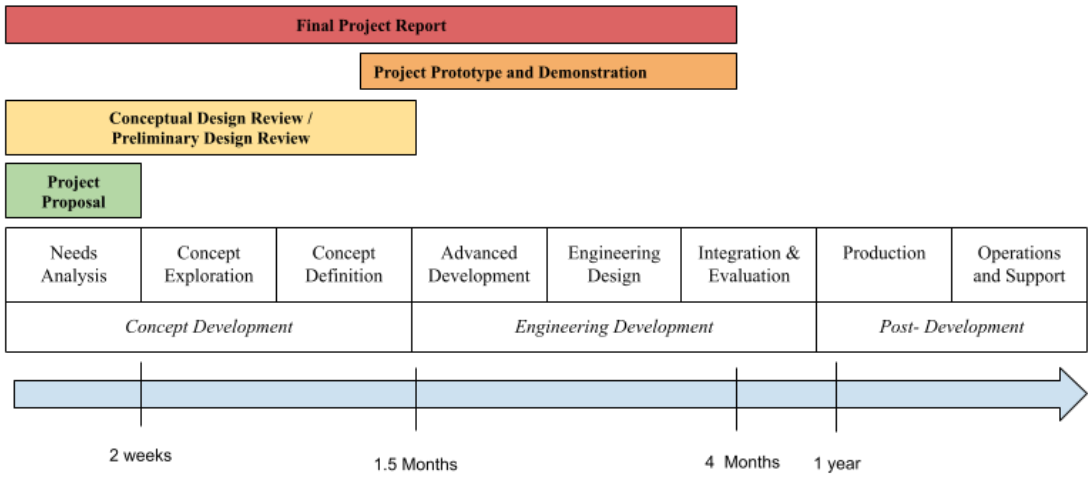
Customers and Users support Development, Operational, and Acceptance phases. They work with Design Engineers during development and provide honest feedback during testing. They also lead Acceptance testing with help from the Systems Engineers.

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10.2.1 Schedule

T&E objectives will be satisfied throughout HPC development.

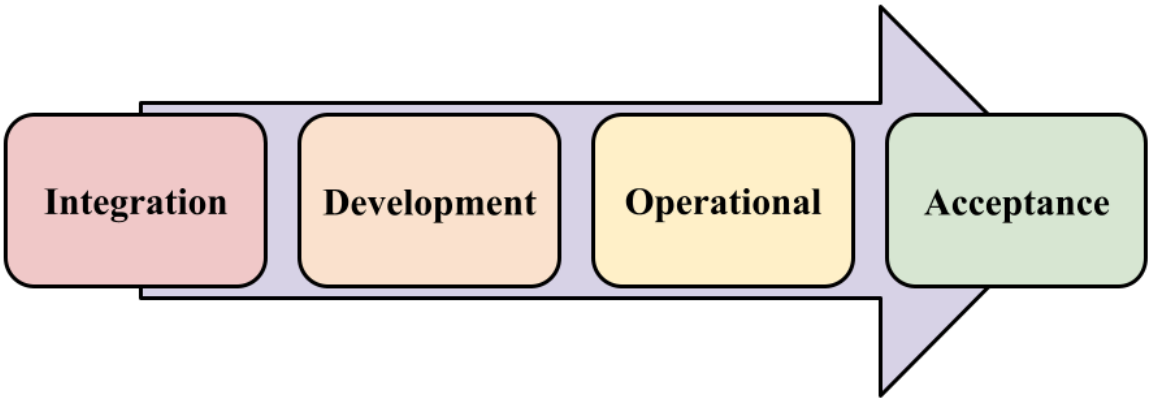
The timeline below illustrates when each objective will be performed.



Concept development to the prototype Demonstration will extend 4 months The blue arrow underscores the taks and objectives during the timeframe. The 1 year line indicates the possible end of testing and evaluating before full production if we decide to proceed to attempt to make the HPC a real product.

T&E phases will be executed linearly in the Integration & Evaluation block/timeframe. These phases include Integration, Development,

Operational, and Acceptance T&E which are discussed further in section 10.3



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10.3 Test and Evaluation Strategy

10.3.1 Integration Testing and Evaluation

Integration testing focuses on combining the Power System, Sensory System, and Physical System into one working device. Design Engineers lead this phase. Testing takes place in a lab environment approved by the Systems Engineers.

Integration testing checks that all mechanical and electrical interfaces function correctly. It verifies that sensors receive power, that the generator output flows through the regulation circuitry, and that the display receives correct data from the Arduino. Simulated loads and controlled cranking speeds are used to confirm expected performance.

10.3.2 Development Testing and Evaluation

Development testing focuses on verifying that the HPC meets its requirements. Design Engineers lead this phase, and the Operational Requirements in section 4 guide the test creation. This includes electrical output verification, thermal testing, mechanical stress testing, display accuracy, and data update rate checks.

Results are compared to predictions and requirements. All findings are reviewed by SEs, specialists, and users.

10.3.3 Operational Testing and Evaluation

Operational testing focuses on validating that the HPC works in realistic conditions. Independent Testers lead this phase with support from SEs and DEs. Testing occurs in practical use environments such as outdoor locations, warm rooms, and scenarios where the user charges an actual phone.

This phase checks ease of use, long duration cranking, durability after real world drops, and safe surface temperatures. A readiness review takes place at the end of the phase.

10.3.4 Acceptance Testing and Evaluation

Acceptance testing is led by customers and users, with Systems Engineers and specialists providing support. The HPC is tested in fully operational form. The tests include functional tests, power delivery tests, display accuracy, and scenario based evaluations like charging a dead phone from zero percent.

Results are presented in a final readiness review.

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10.3.5 Approaches

Testing uses both single event tests and scenario based tests. Single event tests evaluate one function at a time, like voltage stability or RPM reading accuracy. Scenarios combine multiple steps, such as dropping the device, picking it up, and immediately cranking to test survivability and functionality.

Below are three example HPC test events.

Test Event 1: Crank Input to Electrical Output Verification

- Tester: Systems Engineer
- Participants: DEs, Specialists, DA, Users
- Subsystems: Power System, Sensory System
- Objective: Verify that cranking at 80 RPM produces at least 4000 mA at 5 volts through the onboard regulation system.
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- Constraint: Must use a standardized load.
- Pass Criteria: Voltage stays within 5 percent and current meets requirement.

Procedure:

1. Attach standard USB load.
2. Begin cranking at 80 RPM.
3. Record voltage and current.
4. Confirm output meets specs.

Test Event 2: Display and Sensor Accuracy Test

- Tester: Data Analyst
- Participants: SEs, DEs, Users
- Objective: Confirm the display updates at least 4 times per second and shows correct voltage and RPM values.
- Constraint: Arduino and sensors must be powered by the Power System.
- Pass Criteria: Accuracy within tolerance defined in section 4.

Procedure:

1. Apply known test voltages.
2. Crank at measured RPM levels.
3. Compare sensor readings to reference instruments.

Test Event 3: Drop and Durability Validation

- Tester: Independent Tester
- Participants: SEs, DEs, Specialists
- Objective: Confirm the HPC survives three drops from one meter and remains functional.
- Constraint: Full system must be assembled.
- Pass Criteria: No major damage and device still operates.

Procedure:

1. Drop HPC in each orientation.
2. Inspect device.
3. Attempt to crank and charge a test load.

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10.4 Test and Evaluation Limitations and Constraints

The HPC faces several environmental and physical restrictions. These include heat buildup from the generator, user fatigue during long cranking sessions, and risks from dust or moisture in outdoor environments. The HPC must operate within 40 to 100 degrees Fahrenheit, and its surfaces cannot exceed 48 degrees Celsius during use. This limits when and where testing can occur. Electrical interference is minimal in comparison to radio based systems, but noise in the generator's output and sensor wiring can affect readings. Proper grounding and shielding must be maintained throughout testing. Lastly, all T and E efforts must follow safety requirements for rotating hardware, pinch points, and handling heated surfaces.

10.5 Test and Evaluation Summary

Testing the HPC requires measurement tools for voltage, current, power, RPM, thermal data, and structural integrity. The following equipment is typically needed:

- Multimeters
- Oscilloscopes
- Temperature probes
- Tachometers
- USB electronic loads
- Drop testing surfaces
- Crank speed fixtures
- Logging equipment
- Display monitoring tools

All T and E phases must meet exit criteria established by Systems Engineers and Data Analysts. Once all phases are complete and requirements are met, the HPC is cleared for acceptance.

11. Summary of Advantages and Strengths of the System Design

The HPC delivers a reliable, durable, and informative emergency charging system. By combining a strong mechanical design with accurate real time sensor feedback, the HPC gives users confidence that their effort is producing valuable power. The Sensory System enhances the experience by showing voltage, RPM, power output, and system behavior, which helps users crank more efficiently.The enclosure materials and mechanical layout allow the HPC to survive drops, side loads, and general day to day handling. The crank mechanism is designed to be comfortable and consistent, helping minimize fatigue while maximizing power input.The regulated 5 volt output, current protection, and stable USB connection make the HPC safe for phones and small electronics. The internal layout ensures heat does not damage components or cause discomfort to the user. Overall, the HPC provides a strong advantage compared to commercial hand crank chargers. It delivers more real time feedback, better structural durability, higher output consistency, and clearer user interaction. It stands out as a student engineered system that brings together the best features of existing products and improves on the areas where they fall short.

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13. RACI MATRIX

RACI MATRIX Responsible | Accountable | Consulted | Informed

RACI Matrix — Team Members: Ebube, Thinh, Steven, Joshua					
Objective / Task	**Ebube**	**Thinh**	**Steven**	**Joshua**	**Task Description (What They Do)**
Circuit Modeling	R	A	C	I	E: Designs and simulates circuit layouts. T: Oversees quality and ensures alignment with system goals. S: Provides input on component feasibility. J: Informed on progress and results.
HPC Objective and CONOPS	C	A	R	I	E: Advises on computational requirements. T: Defines high-level objectives and operational concepts. S: Implements setup and testing. J: Updated the overall plan.
System Context Diagram	A	R	C	I	E: Approves final diagram. T: Creates and manages diagram structure. S: Consulted for technical relationships. J: Informed of changes.
Operational Requirements	C	R	A	I	E: Provides engineering constraints. T: Writes operational criteria. S: Ensures requirements align with system performance. J: Notified of updates.
Functional Description	A	R	C	I	E: Approves system functions documentation. T: Writes and organises functional flow. S: Consulted for subsystem inputs. J: Informed of progress.

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Physical Description	R	C	A	I	E: Drafts the physical layout. T: Gives feedback on design practicality. S: Finalises and approves physical specifications. J: Informed of design completion.
Research	R	A	C	I	E: Conducts primary research on design and materials. T: Guides direction and validates sources. S: Consulted for application relevance. J: Receives summaries.
Free Gather (Data/Info)	C	R	A	I	E: Consulted on what data to collect. T: Manages and organises data collection. S: Gather information and documentation. J: Updated on findings.
Getting Parts	I	C	R	A	E: Notified of required materials. T: Verifies part compatibility. S: Procures and checks all components. J: Oversees completion and approves purchase.
Physical Modeling	R	C	A	I	E: Builds and tests prototype models. T: Consulted for integration. S: Leads assembly and ensures functionality. J: Informed on testing results.
Assembling	C	I	R	A	E: Offers assembly guidance. T: Updated on progress. S: Assembles all components. J: Oversees and ensures build completion.
Edit Report	A	C	I	R	E: Reviews and approves the final version.

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					T: Provides feedback on technical content. S: Informed of final submission. J: Edits, formats, and finalises the report.

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